Cambridge Assessment International Education

Cambridge IGCSE™ (9–1)

CANDIDATE NAME	

CENTRE NUMBER						CANDIDATE NUMBER					

PHYSICS **0972/41**

Paper 4 Theory (Extended) **May/June 2022**

1 hour 15 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages.

DC (EF) 311602
© UCLES 2022

www.root19.com +91 - 9969 353 391

1 A car of mass *m* is travelling along a straight, horizontal road at a constant speed *v*.

At time *t* = 0, the driver of the car sees an obstruction in the road ahead of the car and applies the brakes.

The car does **not** begin to decelerate at *t* = 0.

(a) Explain what is meant by deceleration.

.....

.....

..... [2]

(b) Suggest **one** reason why the car does **not** begin to decelerate at *t* = 0.

.....

..... [1]

(c) Fig. 1.1 is the distance–time graph for the car from *t* = 0.

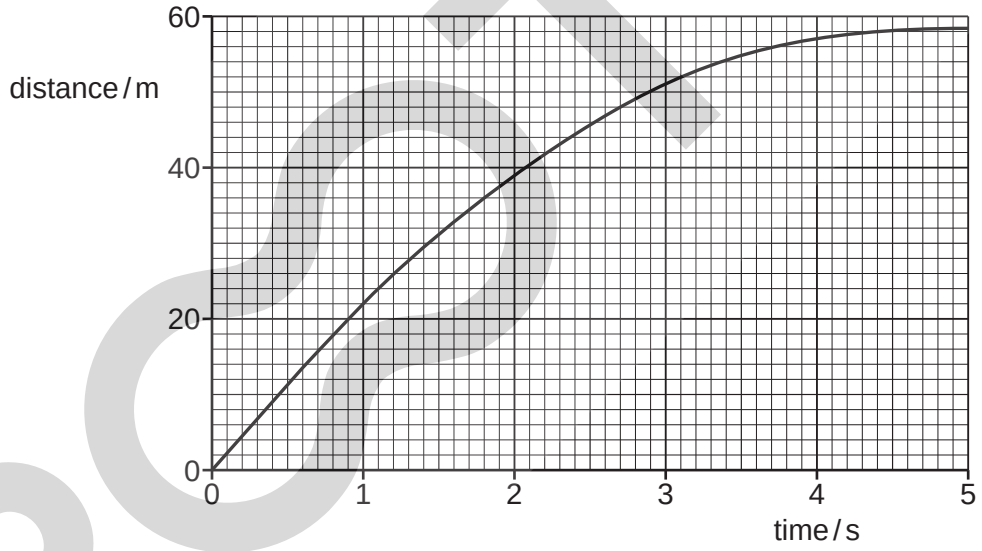

Fig. 1.1

(i) State the property of a distance–time graph that corresponds to speed.

..... [1]

(ii) Using Fig. 1.1, determine the initial speed *v* of the car.

v = ... [2]

www.root19.com +91 - 9969 353 391

(d) When the car is decelerating, there is a constant resistive force F on the car due to the brakes.

The deceleration of the car is greater than $\frac{F}{m}$ and is **not** constant.

Explain why:

(i) the deceleration of the car is greater than $\frac{F}{m}$

.....

..... [1]

(ii) the deceleration is **not** constant.

.....

.....

..... [2]

[Total: 9]

ROOT 19

www.root19.com +91 - 9969 353 301

2 Fig. 2.1 shows water stored in a reservoir behind a hydroelectric dam.

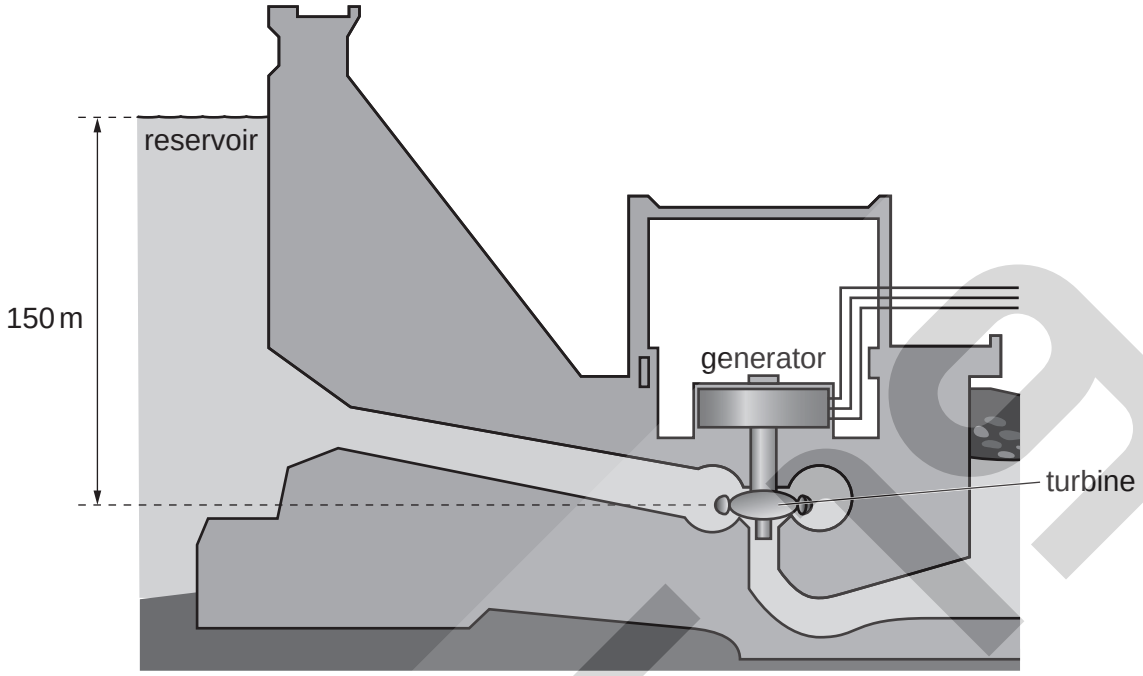

Fig. 2.1 (not to scale)

(a) State the form of the energy stored in the water in the reservoir that is used to generate electricity.

..... [1]

(b) The turbine is 150 m below the level of the water in the reservoir.

Atmospheric pressure is 1.0×10^5 Pa. The density of water is 1000 kg / m^3.

(i) Calculate the total pressure in the water at the turbine.

pressure = ... [3]

www.root19.com +91 - 9969 353 391

(ii) The turbine has a cross-sectional area of $3.5\,m^2$.

Calculate the force exerted on the turbine by the water.

force = ... [2]

(c) The water flows to the turbine through a pipe of constant cross-sectional area.

Explain why the kinetic energy of the water in the pipe remains constant as it flows through the pipe.

.....

.....

..... [2]

[Total: 8]

© UCLES 2022

www.root19.com +91 - 9969 353 301

3 During a picnic on a warm, dry day, a metal can of lemonade is wrapped in a damp cloth.

Evaporation cools the water in the cloth.

(a) Explain, in terms of molecules, how evaporation cools the water in the cloth.

.....

.....

.....

..... [3]

(b) As the water in the cloth cools, so does the lemonade.

Explain how electrons transfer thermal energy through the metal of the can.

.....

.....

.....

..... [3]

[Total: 6]

© UCLES 2022
www.root19.com

+91 - 9969 353 391

4 A thermocouple is a device that is used as a thermometer.

(a) Fig. 4.1 shows a beaker that contains molten sulfur at an initial temperature greater than 400 °C.

(i) On Fig. 4.1, sketch and label a diagram of a thermocouple that is used to determine the temperature of the sulfur as it cools to room temperature.

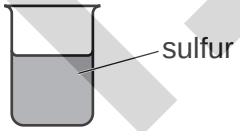

Fig. 4.1

[4]

(ii) Describe briefly how the temperature of the sulfur in the beaker is deduced.

.....

.....

..... [2]

(b) State **one** advantage of using a thermocouple to measure temperature rather than using a liquid-in-glass thermometer.

.....

..... [1]

[Total: 7]

© UCLES 2022

www.root19.com
+91 - 9969 353 301

5 Fig. 5.1 shows a kitchen tap that supplies instant boiling water.

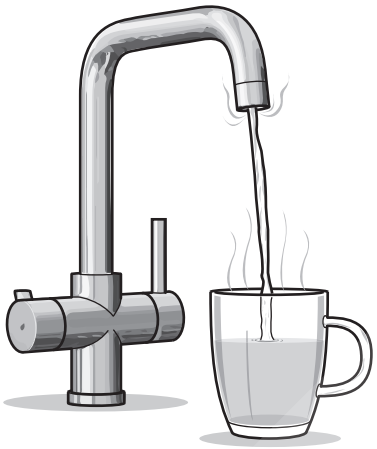

Fig. 5.1

Cold water passes over an electric immersion heater inside the tap.

The boiling point of water is 100 °C.

(a) State what is meant by boiling point.

.....

..... [2]

(b) The immersion heater is powered by the mains at a voltage of 230 V. When the tap is opened, the heater switches on and the current in the heater is 13 A.

(i) Calculate the thermal energy produced by the heater in 60 s.

thermal energy = ... [2]

(ii) The specific heat capacity of water is 4200 J/(kg °C). The cold water that enters the tap is at 22 °C.

Calculate the rate at which water at its boiling point emerges from the tap.

rate = ... [4]

www.root19.com +91 - 9969 353 391

(c) The metal tap is earthed and there is a fuse in the cable that connects the heater to the mains.

1. Explain how the earth wire protects the user.

.....

.....

2. Explain how the fuse protects the circuit.

.....

.....
[3]

[Total: 11]

ROOT 19

www.root19.com +91 - 9969 353 301

6 Fig. 6.1 shows a road next to the sea.

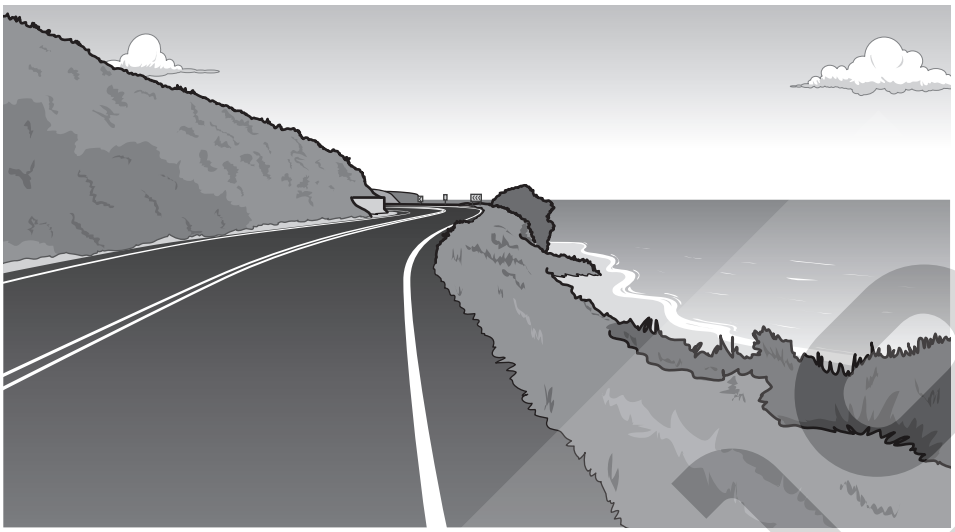

Fig. 6.1

(a) On a sunny day, the Sun warms the road.

Describe how energy from the Sun reaches the Earth and warms the road.

.....

.....

.....

..... [3]

(b) The temperature of the road is greater than the temperature of the sea.

The surface of the road is black.

Suggest **one** reason why the temperature of the road is greater than that of the sea.

.....

..... [1]

(c) The air above the road is heated by the warm road.

(i) Describe how this affects the molecules of the air.

.....

.....

..... [2]

www.root19.com +91 - 9969 353 391

(ii) A cyclist travelling along the road notices that a cool breeze is blowing from the sea to the land.

Explain how convection produces this breeze. You may include a diagram if it helps your answer.

.....

.....

.....

..... [3]

[Total: 9]

www.root19.com +91 - 9969 353 301

7 Fig. 7.1 is a full-scale diagram of a small nail N in front of a thin converging lens. The line L represents the lens.

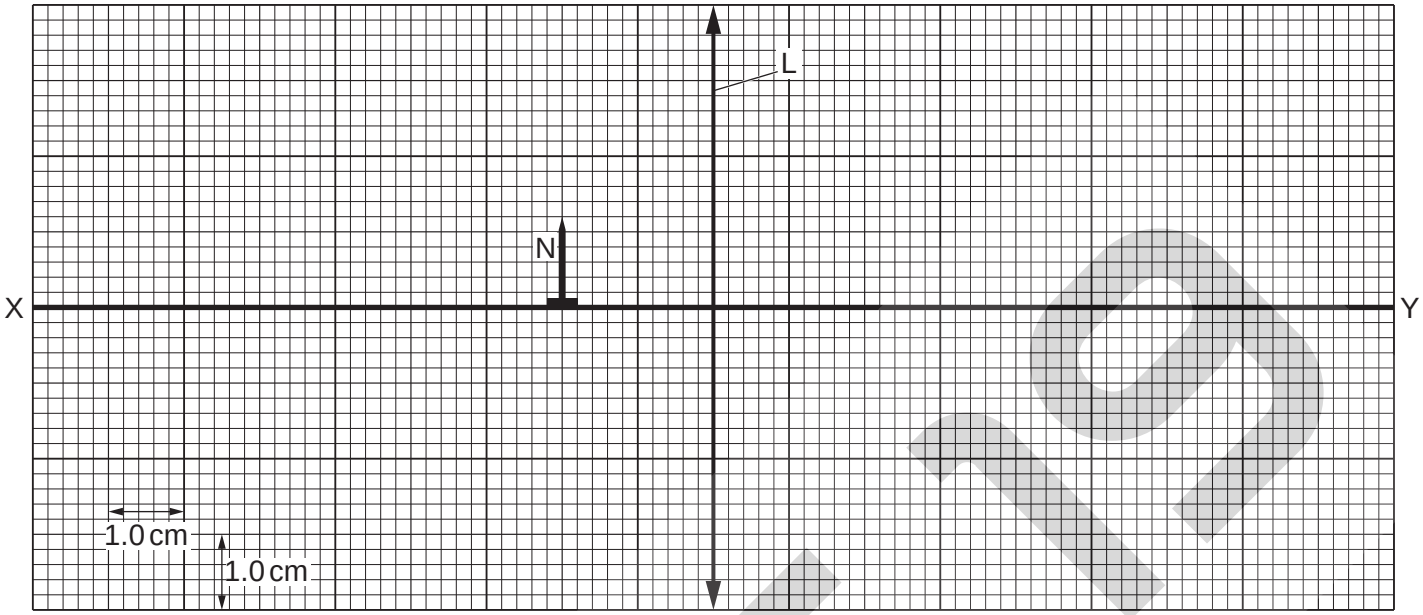

Fig. 7.1 (full scale)

The focal length of the lens is 3.0 cm.

(a) Rays of light, parallel to XY, are travelling towards the lens.

Describe what happens to the light after it passes through the lens.

.....

.....

.....

..... [3]

(b) On Fig. 7.1, mark and label with an F each of the **two** principal focuses of the lens. [1]

(c) The small nail N, of height 1.2 cm, is positioned 2.0 cm to the left of the lens.

(i) By drawing on Fig. 7.1, find the position of the image I of N and add image I to the diagram. [3]

(ii) State and explain whether I is a real or a virtual image.

.....

..... [1]

(iii) State the name given to a lens when it is used in this way.

..... [1]

[Total: 9]

www.root19.com +91 - 9969 353 391

8 Fig. 8.1 shows two vertical, cylindrical tubes and a cylindrical magnet all held in a vacuum.

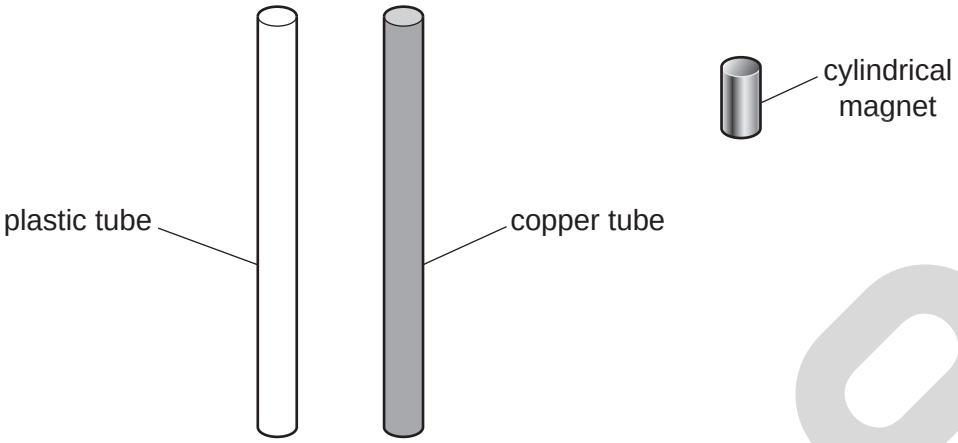

Fig. 8.1 (not to scale)

One tube is made of plastic and the other tube is made of copper. The two cylindrical tubes have identical dimensions.

The magnetic field of the small, cylindrical magnet is extremely strong.

Initially, the magnet is at rest at the top of the plastic tube.

The magnet is released and it falls through the plastic tube without experiencing a resistive force. The magnet takes 0.67 s to fall to the lower end of the plastic tube.

(a) The mass of the magnet is 0.012 kg.

Calculate the kinetic energy of the magnet when it reaches the lower end of the plastic tube.

kinetic energy = ... [4]

(b) The magnet is then held at the top of the copper tube and released. As it falls through the copper tube, an electric current is generated in the copper.

(i) Explain why there is a current in the copper.

.....

.....

..... [2]

www.root19.com +91 - 9969 353 301

(ii) The current in the copper produces a magnetic field of its own in the tube.

The magnet falls much more slowly in the copper tube than in the plastic tube.

Explain why the magnet falls more slowly in the copper tube.

.....

.....

..... [2]

[Total: 8]

© UCLES 2022
www.root19.com

+91 - 9969 353 391

9 Combinations of logic gates are used when digital signals are processed.

(a) Describe the difference between a digital signal and an analogue signal. You may include a diagram if it helps your answer.

.....

.....

..... [2]

(b) Fig. 9.1 is the truth table for a logic gate X.

input A	input B	output
0	0	1
0	1	0
1	0	0
1	1	0

Fig. 9.1

State the name of logic gate X and draw the symbol that represents it.

name .....

symbol

[1]

(c) Logic gate Y is identical to logic gate X.

Draw a combination of logic gates X and Y that behaves like an OR gate. Label the inputs A and B and label the output Q.

[2]

[Total: 5]

www.root19.com +91 - 9969 353 301

10 Two of the isotopes of hydrogen are hydrogen-2 ($^{2}_{1}H$) and hydrogen-3 ($^{3}_{1}H$).

(a) **(i)** State **one** similarity in the composition of their nuclei.

..... [1]

(ii) Describe how a nucleus of hydrogen-3 differs from a nucleus of hydrogen-2.

.....

..... [2]

(b) In a nuclear fusion reactor, a nucleus of hydrogen-2 fuses with a nucleus of hydrogen-3 at an extremely high temperature. This fusion reaction produces an isotope of element X and releases a neutron.

(i) Explain why an extremely high temperature is needed when forcing these two nuclei together.

.....

.....

.....

..... [3]

(ii) Using nuclide notation, complete the equation for this reaction.

$$^{2}_{1}H + ^{3}_{1}H \longrightarrow$$

[2]

[Total: 8]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.